

Exam Project Part 1 – Supervised and Unsupervised Learning

The following gives you an opportunity to work on applying machine learning algorithms. It gives you an opportunity to take a project idea and work through all the stages of the machine learning process, from business case to deployment of a model.

DUE DATE: Upload report to Wiseflow 27 May with exam on 2nd 3rd or 4th June

This is part 1 of the exam project. Steffen will hand out part 2 which is a comparative analysis of this solution and AI generated solutions.

The exam will be 10 minutes to present your solution to parts 1 and 2 of the exam project and then 15 minutes of questions.

Task

Select and prepare a data set to explore a problem you are interested in understanding. Please discuss with me what your idea is.

Using appropriate models (such as Random Forest, K Nearest Neighbour, KMeans and DBSCAN) prepare a solution. You may find it useful to combine a kmeans or other clustering approach with a supervised learning model for prediction, as is common when using Principal Components Analysis to reduce the number of dimensions to be explored. If using AI tools to help create your solution you must document this as part of your solution. If using AI to help create, train and interpret your model you will need to be able to explain what it did at the exam.

You should demonstrate that you have used the data in an appropriate way (e.g. training it in supervised learning) explain your understanding of the process and the results. The focus should be on the data handling process, selection of models, training where needed, interpretation of results and deployment.

In the exam you will have a chance to present your work, answer questions about your project and also more generally about supervised and unsupervised learning.

Deliverables

- A Jupyter notebook or other easy to follow approach containing all the code used with comments/Markdown to explain what you did which you turn into a PDF file for upload. There should be commentary about the process you used and reflections on the project (problems with overfitting for example).
- For the exam you will need a 10 minute presentation explaining and reflecting on the project but you will not need to hand in the presentation. Part of your presentation should include a discussion of your code.

